



Department of Computer Science
Quaid-i-Azam University,
Islamabad, Pakistan

Course Title: Operating Systems (CS-226)

Course Instructor: Dr. Umer Rashid

Course Introduction: An operating system is a crucial and essential part of the system. It is a piece of code/software that must be executed on a machine before any other type of software can be executed. It provides an abstraction between user applications, programs, and machines. An operating system is a program that facilitates the user programs. The essential tasks of an operating system include, but are not limited to, process management, memory management, and storage management. The objective of the course is to develop a basic theoretical understanding of the operating system concepts, along with practical experience.

Course Outline: Course is distributed into following modules.

Module 1: Basic Understanding of the Operating System

- Introduction to the Operating System.
- Basics of Computer System Organization and Architecture.
- Operating System Structures and Operations
- Process, Memory, Storage management
- Protection and Security.
- Distributed and Special Purpose Systems, Computing Environments.
- Operating System Services; Interface; Call Types; and Programs.
- Operating System Design and implementation; Virtual Machines.
- Operating System Generations.
- System Calls and APIs.

Module 2: Process Management and Scheduling

- Process Concept, Scheduling, and Operations.
- Inter-process Communication and Client-Server Communication.
- Overview of the Multithreading and Multithreading Models.
- Process Scheduling; Scheduling Criteria and Algorithms.
- Client Server, RPC, LPC and multithreading.

Module 3: Process Coordination and Deadlock Handling

- Background of Process Synchronization.
- Critical Section Problem; and Semaphores.
- Monitors and Atomic Transaction.
- Process Deadlocks; Deadlock Characterization and Handling.
- Process Deadlock Prevention, Avoidance, Detection, and Recovery.

Module 4: Memory Management

- Storage Hierarchy and Storage Management Strategies.
- Storage Allocation Continuous, Non-Continuous.
- Fixed Partition and Variable Multiprogramming.
- Storage Placement Strategies.
- Multiprogramming with Storage Swapping.

Module 5: Virtual Memory Management

- Virtual Memory Introduction and Virtual Memory Concept.

- Evolution of the Storage Organization.
- Multilevel Storage Organization.
- Block Mapping; Paging and Segmentation.
- Paging/Segmentation Systems.
- Virtual Storage Management and Page Replacement Strategies.
- Locality; Working Sets.
- Page Fault Frequency Page Replacement; and Demand Paging.
- Page Release; Page Size and Program Behaviour under Paging.
- Disk Scheduling.

Assessment Criteria

Dues	Marks Distribution
Quizzes	10
Assignments	15
Midterm Exam	25
Final Exam	50

Recommended Readings

- Abraham Silberschatz et al., Operating System Concepts, 10th edition.
- H.M. Deitel, Operating Systems, 3rd edition.
- Selected topics and programs from online resources.

Rules & Regulation

- Audio/Visual recording of lectures is prohibited.
- Class timings will be strictly observed.
- Late submitted assignments will not be entertained.
- Quiz any time, any day in pre-scheduled classes.
- Cheating will not be tolerated in any case.
- Attendance must be > 80 % for terminal

S. No.	CLO	BT Level	PLO
CLO1	Explain the Operating System concepts and issues associated with implementation	C2	PLO2
CLO2	Evaluate various types of operating system algorithms and highlight performance issues	C5	PLO3
CLO3	Implement various operating system functionalities by employing modern tools	C3	PLO4, PLO5, PLO6